EdaphNet v2: Complete Project Summary

Checking soil laboratory reports with a cheap independent observation of the sample · status on 3 October 2026 · paper draft v0.2 · repository github.com/Sudeep-Sharma1/edaphnet-v2 (private)

In one paragraph. Soil test reports are sometimes wrong: a report can belong to another sample (a **swap**), or one number can be wrong (a **value error**, such as a ×10 slip). EdaphNet asks whether a cheap, independent look at the same soil, mainly its **colour** as a phone could record it, can tell a laboratory **which reports to re-test** and **what went wrong**. We built a model that checks a whole lab batch at once and tested it, with rules fixed in advance, on 23,219 real soils with planted errors. **Result:** colour helps find swaps (+0.10) but not wrong values (−0.08), so overall it did not beat the best existing method (the main pre-registered claim failed); a lab spectrometer did beat it (+0.09); and the model names the right error type 97–100% of the time. The work is publishable as an honest study of *when a cheap witness is enough*.

1. Problem and motivation

Soil reports drive fertiliser, liming and carbon decisions. Errors happen in two ways:

- **Swaps:** samples are mislabelled or placed in the wrong tray position, so a farmer receives another soil's results. A swapped report looks perfectly normal, because it is a real report of a different soil, so lab-only checks cannot see it.
- **Value errors:** decimal slips (×10, ÷10), unit mix-ups, swapped digits (0.16 → 0.61), swapped columns, or analytical failures.

Measurement disagreement is not rare: in one comparison of eight service labs, 28% of soil organic carbon results fell outside the proficiency-testing acceptable range (cited in the draft; to be verified). Labs control quality with reference samples and rules, but an individual farmer has no independent way to check *their* report. Re-testing everything is too expensive, so labs need to know **which few reports to re-test**.

2. Research objectives and questions

1. **Detection:** rank reports so that re-testing the top 5% catches as many errors as possible.
2. **Attribution:** say what went wrong (which sample it was swapped with, or which value and which kind of slip).
3. **Correction:** propose the corrected value or the correct sample.
4. **Detectability:** map which errors each witness strength can reveal (none → colour → spectrometer).
5. **Error control (secondary):** flag reports with a guaranteed false-alarm rate (FDR 10%).

How the research question changed (all changes recorded)

Version	Claim	Outcome
v1 (doc 11)	A phone-grade colour witness catches more errors at a guaranteed 10% false-alarm rate	Failed (doc 17), partly due to our calibration mistake
v2 (doc 18, pre-registered)	With a 5% re-test budget, the method finds more errors than the best existing method, using colour	Failed on new data (KSSL, doc 21)
Current framing	When is a cheap witness enough? Detectability, attribution, calibration requirements	Supported by the evidence

3. Existing work

A targeted search of papers, patents and products (docs 09, 10) compared 16 prior works against the elements of our method.

Area	What exists	Why it is not our problem
Soil properties from phone images	Many papers predict organic carbon, texture, iron, EC or Munsell colour from photos; LandPKS app; a 2026 open dataset of 84 samples × 4 phones	They *predict* properties; they do not check lab reports
Checking lab values against predictions	Process-industry patent (US 8,352,394); clinical autoverification and ML "delta checks" for wrong-blood-in-tube	Use patient history or process sensors, not an independent look at the same specimen
Sample-swap detection	Genomics: Broman et al. 2014, Crosscheck (matching two data types per person)	Need near-perfect genetic witnesses; no value errors
Numeric error detection	cleanlab regression label-error scores (Zhou et al. 2023)	General tool; used as baseline B7
Conformal outlier testing	Conformal p-values + Benjamini–Hochberg (Bates et al. 2023; Jin & Candès 2023)	General tool; used as a component
Soil spectroscopy QC	Spectra flag wet-lab inconsistencies (KSSL); uncertainty-based rejection sends bad *predictions* back to the lab	Uses lab spectrometers; reverse direction
Practice	LUCAS photographs each sample so it can be checked manually; lab software uses rules and barcodes	Manual or preventive, not statistical detection

4. Research gaps

- No method found that uses an **independent observation of the same soil sample** to **jointly** detect, explain and correct **swaps and value errors across a lab batch**.
- No **detectability map**: which report errors a witness of a given strength can reveal.
- No **public benchmark** of soil lab-report errors (LimeSoDa and REIN are the nearest; neither has lab errors).
- No study of whether **conformal false-alarm control** stays valid with structured lab errors and realistic calibration data.

The "no prior work" statements still need the formal Scopus / Web of Science search (planned; HANDOVER.md step 3).

5. Proposed methodology

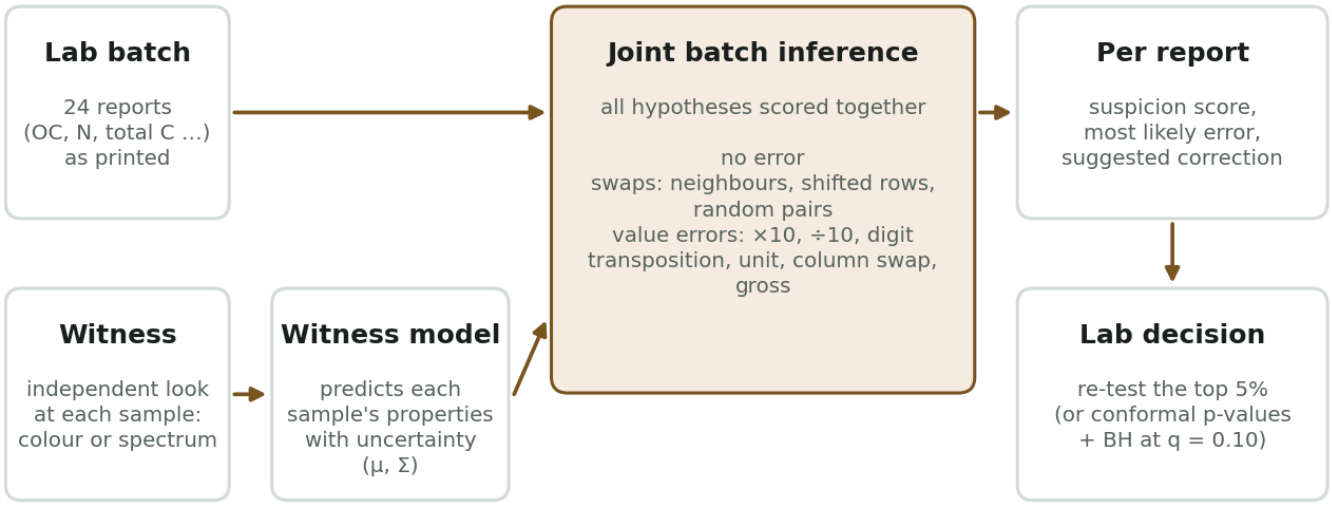


Figure 1 of the paper. The method in one picture.

1. **Witness model.** Each property is put on a standard scale (log or logit where needed). From the witness (colour, 10 visible bands, or the full spectrum) a gradient-boosted model predicts each property; the leftover errors give a full covariance, so the model knows how carbon, nitrogen and total carbon move together. Result: for each sample, a predicted distribution of its properties.
2. **Value-error library.** ×10, ÷10, unit confusions, swapped adjacent digits of the printed number, swapped columns, and a gross multiplicative error over a grid. At most one corrupted value per report.
3. **Swap hypotheses.** No swap; neighbours swapped; whole tray shifted by one or two places (rotations); any two samples swapped.
4. **Joint batch inference.** For every report and every candidate sample, the model scores how well the printed report fits that sample's witness under each possible error. All swap hypotheses are scored together over the batch, exactly over the enumerated set.
5. **Output per report.** Suspicion score = 1 − probability the report is clean; the most likely explanation (swap partner or error type); a proposed correction.
6. **Lab decision.** Re-test the top 5% (primary). For error control: conformal p-values against clean calibration reports, then Benjamini–Hochberg at q = 0.10, using **one score per distinct calibration sample**.

6. Datasets

Dataset	Samples	Properties	Licence	Role
Synthetic (T0)	generated	8, with a colour-like witness of chosen strength	—	Test the mechanism (doc 13)
ICRAF-ISRIC (OSSL v1.2)	3,473 (553 profiles)	pH, OC, clay, sand, CEC, K, Ca, Mg	CC-BY	First real test (doc 17)
KSSL Vis-NIR (OSSL v1.2)	19,746 (6,999 locations)	OC, N, total C	CC-BY	**Pre-registered test** (doc 21)
LUCAS topsoil	≈ 19,000 (2018)	pH, OC, N, P, K, $CaCO_3$, texture …	on request	Planned confirmation (not used)

Own photos + lab reports (step 6)	planned 200–300	Soil Health Card panel + texture	ours	Paper 2: real photos and real errors

What is real and what is simulated. The soils, lab values and spectra are **real**. The **errors** were planted by our code (a swap event in 20% of batches; value errors in 2.5% of other reports; about 5% erroneous reports in total), because only then do we know where the errors are. The **"phone" colour** was calculated from each lab spectrum, plus camera-like noise (ΔE 0, 2, 4, 8): a best case, since lab soil is dried and ground. Splits are by profile or location (train 50%, calibration 15%, test 35%), so test soils come from places never seen in training. Rebuilding the KSSL dataset from the downloaded files gives an identical result.

7. Models

Model	What it is	Used for
EdaphNet (Full)	Joint batch inference over swaps × error library, using the witness model	The proposed method
Witness models	LightGBM regression per property (300 trees) on CIELAB colour, 10 bands, or 30 spectral PCA components; Gaussian with full residual covariance	Predict properties from the witness
B0 range checks	Value outside the 0.1–99.9% range of clean reports	Baseline
B1 soil rules	Consistency rules, e.g. organic carbon ≤ 1.05 × total carbon	Baseline
B2 lab-only density	How unusual the report is on its own (multivariate)	Baseline
B3 witness interval	Each value checked separately against the witness prediction	Baseline
B4 swap matcher	Optimal reassignment of reports to samples (Hungarian algorithm, genomics-style)	Baseline
B5 supervised	LightGBM classifier trained on planted errors (knows the true error mix: an unfair advantage)	Baseline, usually the strongest
B6 rule-based repair	First unit/decimal conversion that brings all values into range	Attribution baseline
B7 cleanlab	Regression label-quality score (Zhou et al. 2023)	Baseline
Demo colour model	Colour → OC / N / total C for the live demo	Demo only
App leaf model	MobileNetV3-Large features + logistic head, "not sure" below 50% confidence	App only (not trained yet)

8. Experiments

Experiment	Purpose	Run folder (results/runs/)
Simulation (30 settings)	Does the mechanism work, and for which witness strength?	`20260923-230637_sim_t0`
Witness ladders	How much does colour / spectrum know about each property?	`…witness_ladder_icraf8`, `…_kssl3`
ICRAF test (42 settings)	First go/no-go on real soils (pre-registered v1)	`20260925-011337_real_icraf8`

Calibration diagnostic	Why did false-alarm control fail?	`20261001-233621_diag_calibration`
KSSL test (50 settings, 5 seeds)	**Pre-registered test on new data (A1–A4)**	`20261001-234902_real_kssl3_095deaa`
Unknown errors (1 seed, then 5 seeds)	Does the pattern hold for error types the model does not know?	`…mismatch_7099ff1` , `…mismatch5_d54e40c`
Exploratory re-analyses	Swaps vs values, ablations, pool size, gross errors, error mix	`results/exploratory/`

Settings: batch of 24 reports; pools of 50 batches; 5 seeds; random and clustered (same location) batch orders; witness rungs W0 (none), W1 colour at ΔE 0/2/4, W3 spectrometer.

9. Evaluation metrics

Metric	Meaning
Recall at 5% budget (primary)	Share of erroneous reports found when the top 5% most suspicious reports per pool are re-tested
Power at FDR q = 0.10	Share of errors flagged when false alarms are controlled at 10%
Realised FDR	Actual share of false alarms among flagged reports
Attribution accuracy	Share of flagged errors whose type (or swap partner) is named correctly
Detection by error type	Recall for each kind of error and each witness (detectability table)
R^2	How much of a property the witness explains, on unseen locations
95% intervals	Cluster bootstrap over pools (2,000 resamples), paired comparisons on identical batches

10. Results

10.1 What colour knows (R^2 on unseen locations)

KSSL: organic carbon 0.72 (0.70 with phone-like noise), nitrogen 0.55, total carbon 0.69. ICRAF: organic carbon 0.37, pH 0.27, Ca 0.23, clay 0.16, CEC 0.11, K 0.07, sand 0.02. A spectrometer explains 0.39–0.90 of all properties. **Colour mainly knows carbon.**

10.2 Simulation

The method beat all baselines only with a moderately informative witness (e.g. 0.65 vs 0.42 of errors caught at FDR 0.10 when colour explains ~66% of carbon); with a weak witness the gain was ≈ 0. The gain came almost entirely from swaps.

10.3 Pre-registered KSSL test (the main result)

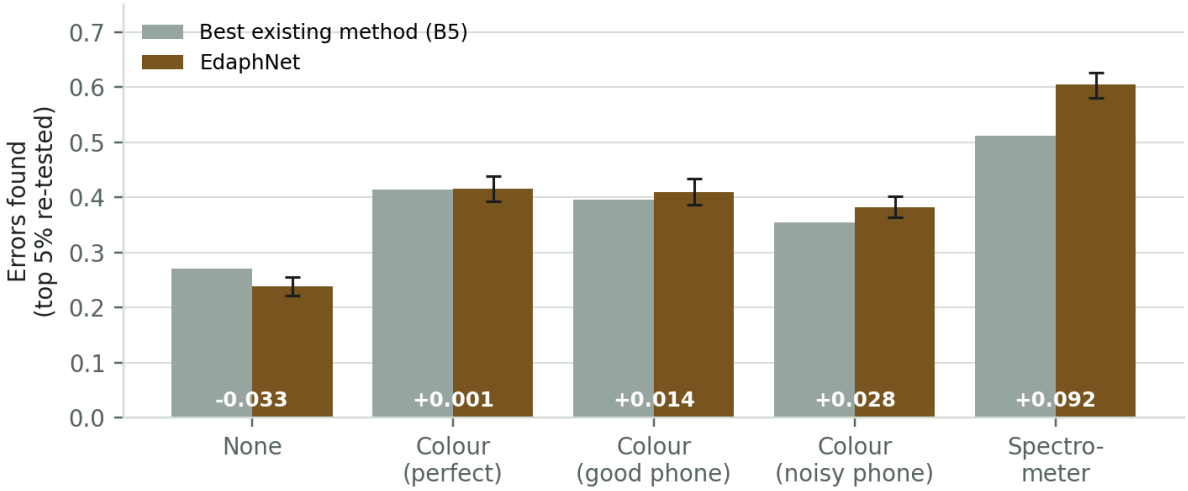


Figure 3 of the paper. Errors found at a 5% re-test budget, EdaphNet vs the best existing method.

Witness	EdaphNet	Best baseline	Difference [95% CI]
None	0.238	0.271	−0.033 [−0.049, −0.016]
Colour, perfect	0.416	0.415	+0.001 [−0.022, 0.023]
Colour, good phone (ΔE 2)	0.410	0.396	+0.014 [−0.008, 0.038]
Colour, noisy phone (ΔE 4)	0.382	0.354	+0.028 [0.010, 0.049]
Spectrometer	0.604	0.512	+0.092 [0.069, 0.113]

Verdict: A1 (≥ +0.05 with colour) failed; A1-c (not worse in clustered batches) passed; A2 (false alarms ≤ 12%) passed, but with colour the method flagged almost nothing at the registered pool size; A3 (attribution beats rule repair by ≥ 0.20) passed.

10.4 Why colour did not win: swaps vs wrong values (exploratory)

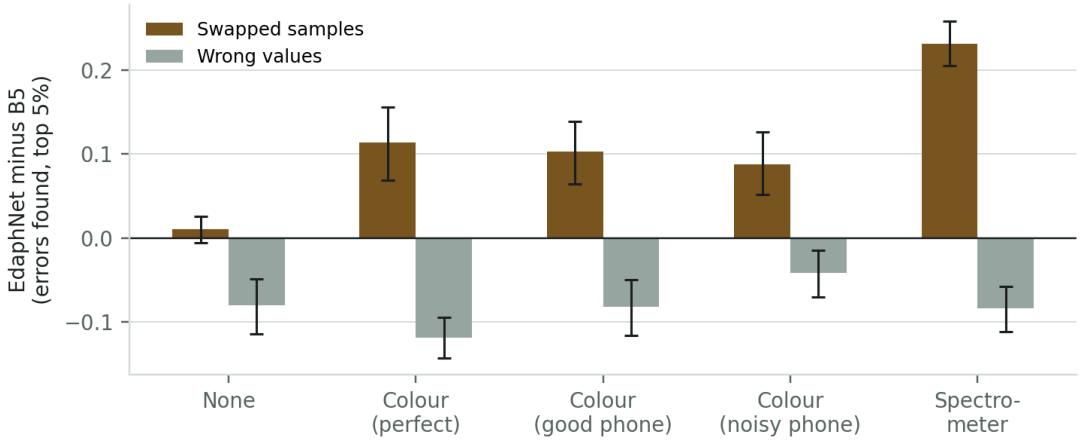


Figure 4 of the paper. EdaphNet minus the best baseline, for swaps and for wrong values.

With a good-phone colour witness, EdaphNet found **+0.103 [0.064, 0.139] more swaps** and **−0.082 [−0.116, −0.050] fewer wrong values** than the supervised baseline; at an equal mix the two cancel. Wrong values were lost mainly on gross errors (0.03 vs 0.26) and ×10 slips (0.68 vs 0.80). The swap hypotheses and the witness create the swap advantage (+0.39, +0.40); the error library trades swap sensitivity (−0.07) for slightly better value detection (+0.02); the structured swap prior had no effect.

10.5 Who wins depends on the error mix (exploratory)

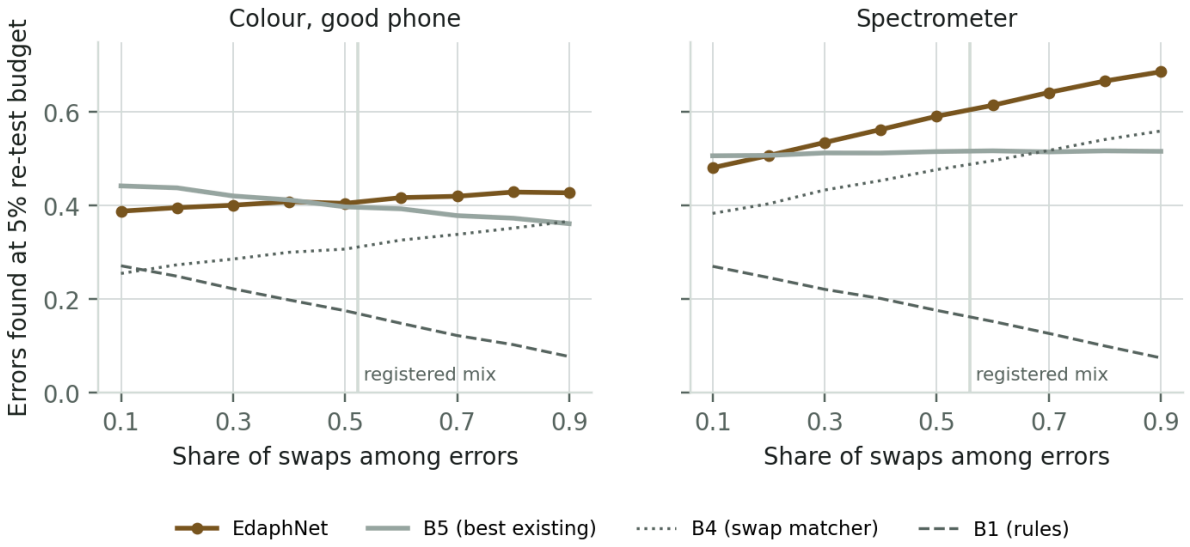


Figure 6 of the paper. Errors found as the share of swaps among errors changes.

Re-ranking the stored scores under other mixes: with colour, EdaphNet beats the best baseline once **60–70% of errors are swaps** (e.g. +0.066 [0.037, 0.095] at 90%) and loses when few are (−0.054 at 10%); with a spectrometer it wins from **30% swaps**. It beats the simple swap matcher at every mix. A lab that knows its own error mix can read off which checker to use.

10.6 Other findings

- **Explaining errors (clearest advantage):** correct error type in 97–100% of flagged value errors (KSSL, spectrometer; pre-registered A3), vs 24–29% for rule-based repair; 89–98% on ICRAF. With colour, too few value errors were flagged to evaluate.
- **Unknown error types (5 seeds):** same pattern: swaps +0.103 [0.069, 0.137], wrong values −0.057 [−0.078, −0.036], overall +0.027 [0.009, 0.047] (colour, random batches).
- **Gross errors:** a real limitation, not a bug. EdaphNet ranks them fairly high but explains them as digit slips, and it does not use the rule "organic carbon ≤ total carbon" that the baselines exploit.
- **False-alarm control:** fails silently when calibration samples are reused (realised FDR 0.57 → 0.00 after de-duplication). Its power depends on calibration size relative to how many reports are tested at once: Benjamini–Hochberg over m reports can reject only when at least m/(q(N+1)) reports reach the smallest possible p-value (≈ 5 for a pool of 1,200, fewer than 1 for one batch). Testing one batch at a time restored power to 0.19–0.25 at FDR ≈ 0.10 (exploratory).
- **Detectability:** swaps go from ~3% found without a witness to 40–55% with colour and up to 87% with a spectrometer; digit transpositions and moderate gross errors stay hard for every method.

11. Hypotheses: status

ID	Hypothesis	Status
H1 / A1	The method beats the best baseline with a colour witness	Failed (ICRAF, then KSSL)
H2	The advantage is largest for the weakest witness	Refuted (gain grows with witness strength)
H3 / A2	False-alarm rate stays controlled	Failed on ICRAF (our calibration mistake); passed on KSSL after the fix

H4 / A3	Better error attribution than rule-based repair	Passed
H5 / A4	Detectability map	Reported descriptively (doc 19); the formal power model was not fitted
H6	The structured swap prior helps	Not supported
A1-c	Not worse in clustered batches	Passed

12. Novel contributions (honest assessment)

Contribution	Status
Joint batch model for swaps + typed value errors driven by an independent witness, with explanations and corrections	New as a combination (no equivalent found); its **detection** gain with colour is not established; its **explanation** advantage is clear
Detectability analysis across witness strengths, including the error-mix map (when colour pays off)	New; exploratory parts must be confirmed on new data
Calibration lesson: reuse of calibration samples silently breaks conformal FDR control; power depends on calibration size vs pool size	Useful methodological finding for report verification
Pre-registered, reproducible evaluation with negative results reported	Strength of the study; not a novelty by itself
Not new (used as tools)	Flagging values via predictions, conformal p-values + BH, cross-modal swap matching, unit-error tables, phone soil colour

13. System architecture

Research pipeline (paper)
OSSL data → datasets with location-disjoint splits → witness ladder → `src/edaphnet2` method + baselines → experiments with run records → analyses → figures → paper

Demo layer
Colour model (`models/`) + the method → `serve_demo.py` (Python standard-library web server) → `live.html` with animated pipeline and guided demo; offline presentation page built from run records

App layer (separate)
`app/server.py` → leaf disease (MobileNetV3), crop suitability (FAO EcoCrop rules), soil colour, report check → 4-tab phone page with camera; every advice item has a source

Engineering choices: CPU only (laptop with 7.8 GB RAM); Python with numpy, pandas, scikit-learn, LightGBM; memory-lean runner that writes results as it goes; fixed seeds; resumable runs that give identical numbers (tested); run records with code version; 34 automatic tests; no web framework (standard-library server).

14. Mistakes made and corrected (kept on record)

Mistake	Correction
Data audit counted empty spectrum rows	Recounted; switched to ICRAF-ISRIC (doc 12 §7)
Calibration reused samples (pseudo-replication), inflating false alarms	One score per distinct sample (D-016/17); doc 17 correction
Runner ran out of memory; seeds not reproducible	Lean runner, stable seeds, resume with equivalence test

Analysis script edited while the KSSL run was in progress	Disclosed in doc 21; thresholds unchanged
Run records labelled every KSSL run "ICRAF-ISRIC"	Label fixed in the script; old records left unchanged; noted in doc 26

15. Limitations

1. Errors were planted, not observed; the real mix of swaps and wrong values is unknown.
2. Colour was calculated from lab spectra of dried soil, a best case for a phone photo.
3. KSSL properties are all carbon-related, which favours colour.
4. B5 was trained on the true planted error mix, an advantage it would not have in practice.
5. Several results are exploratory (not pre-registered) and need confirmation on new data.
6. No Indian soils yet.

16. Development plan

Stage	Work	Status
Steps 1–5c	Prior art, specification, simulation, data audit, witness ladder, ICRAF test, diagnosis, KSSL pre-registered test	Done
Paper	Draft v0.2 with review fixes (gross errors, error mix, 5 seeds, abstract, method diagram)	Done
Reproducibility	Download script, REPRODUCE.md, dataset rebuild verified identical	Done
Demo and app v1	Animated live demo, presentation page; app with leaf, crops, soil, report check	Done (leaf model to train)
Next (student + guide)	Guide review; formal Scopus/WoS search; check citations; LUCAS request; IP office; AI-assistance statement; Zenodo DOI; journal choice; submission	To do
Next study	Register amendment v3 (combined re-test policy: method for swaps + value checker); LUCAS confirmation	Drafted (doc 20)
Paper 2	Real phone photos of Indian soils with lab values (pilot 30–50, target 200–300; e.g. VIT VAIAL lab) and blind duplicates (25% of samples) for real errors	Protocol written (doc 15)
App	Train leaf model (report PlantDoc accuracy); agronomist check of crop rankings; soil type on hold	To do

Suggested journals: SOIL, European Journal of Soil Science, Computers and Electronics in Agriculture.

17. Decisions needed from the guide

1. Submit now (planted errors, calculated colour, limitation stated) or wait about a month to add real photos?
2. Does the paper need Indian soil data, or is India acceptable as the motivation?
3. Agree with the framing "when is a cheap witness enough" rather than a new-method paper?
4. Journal, co-authors, and the AI-assistance statement wording.
5. Patent: consult the IP office before any preprint (the repository was public for a period).
6. Budget and lab partner for step 6.

Every number in this summary comes from the project's documents and run records (docs 13, 14, 17, 19, 21, 23, 26 and the paper draft v0.2). This summary was prepared with the help of an AI assistant (Claude).